

SYSTEMS DESIGN

Systems Analysis & Design

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

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Outline

1 Requirements Engineering

2 Design & Process



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2 Design & Process



Stakeholders Vs. Shareholders

- **Stakeholders** are individuals or groups who have an interest in the success of a project.
- **Stakeholders** can be internal or external to a company. For example, customers, employees, suppliers, and regulators are external stakeholders
- Shareholders are individuals or groups who have an ownership interest in a company.
- Shareholders are internal to a company. For example, investors, owners, and managers are internal stakeholders.

devs
legal
sales



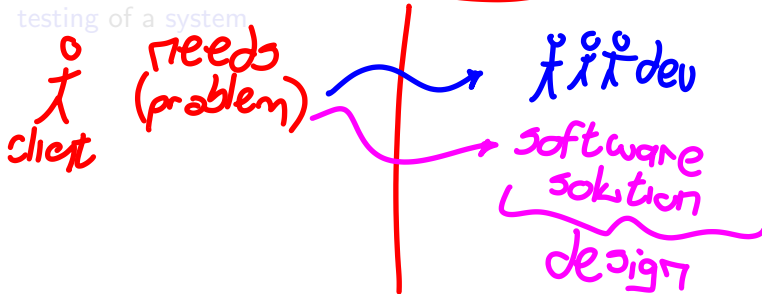
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Requirements

- **Requirements** are statements that describe the features, functions, and constraints of a system.
- **Requirements** are used to communicate the needs of stakeholders to developers.
- Requirements are used to guide the design, development, and testing of a system.



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TDD
Test-driven Development



User Stories

with understanding

- **User stories** are short, simple descriptions of a feature or function of a system. *Expectation*
- They are written from the perspective of the user and describe what the user wants to achieve. *non-technical*
- They are used to capture the requirements of a system in a simple and understandable way.



User Stories: Format Example

User Story

2-Pre-req.

Title:	Priority:	Estimate:
✓ User Story: As a [description of user], <u>I want</u> [functionality] <u>so that</u> [benefit]. Acceptance Criteria: Given [how things begin] When [action taken] Then [outcome of taking action]	🕒	hours effort

role

impact

scenario

test

ProductPlan



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What is Requirements Engineering?

- **Requirements engineering** is the process of eliciting, analyzing, specifying, validating, and managing the requirements of a system. *Handwritten notes: "same page" in red above the word "process"; "details" in red below "specifying"; "same page" in red above "managing".*
- It is a critical activity in the systems development lifecycle that ensures that the system meets the needs of its users. *Handwritten notes: "critical" underlined in blue; "systems development lifecycle" circled in blue; "ensures" underlined in blue; "needs" underlined in blue.*
- It is a collaborative process that involves stakeholders from different backgrounds and perspectives. *Handwritten note: "collaborative process" underlined in blue.*



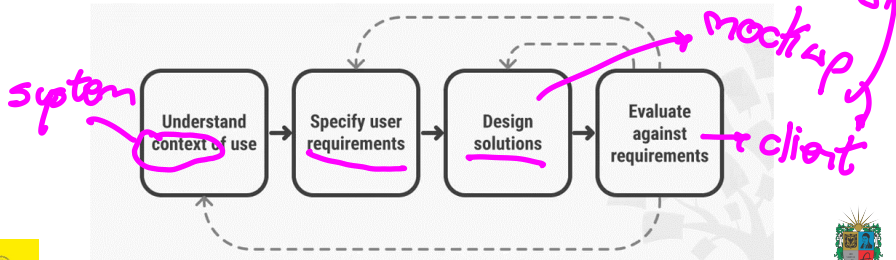
What is Requirements Engineering?

- **Requirements engineering** is the process of eliciting, analyzing, specifying, validating, and managing the requirements of a system.
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User-Centered Design (UCD)

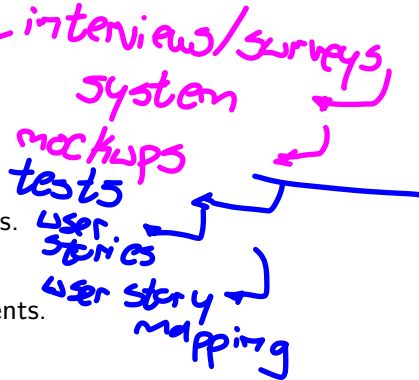
- **User-centered design** (UCD) is an iterative design process that focuses on understanding the needs, preferences, and behaviors of users. *problem!*
- UCD is a collaborative process that involves users in the design and development of a system.
- UCD is used to create systems that are usable, efficient, and satisfying to users. *processes*



Requirements Engineering Process

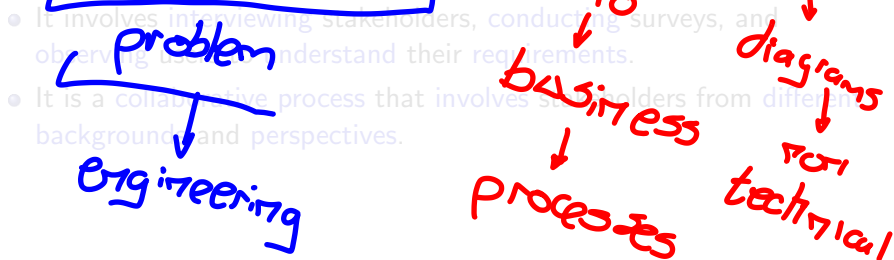
The **requirements engineering** process consists of the following activities:

- **Gathering** requirements.
- Analyzing requirements.
- Validating requirements.
- Verifying requirements.
- Documenting requirements.
- **Managing** requirements.
- Communicating requirements.



Gathering Requirements

- **Gathering** requirements is the process of **collecting** and **documenting** the needs of stakeholders.



Gathering Requirements

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- It involves interviewing stakeholders, conducting surveys, and observing users to understand their requirements.
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align expectations



Clients are not always right

Dear Santa,
 How are you? I'm good.
 Here is what I want for
 Christmas.

A http://www.amazon.com/gp/product/B0032HFG0M/ref=ssq_hps_bw_g21_ir03?pf_rd_m=ATVPDKIKXODER&pf_rd_s=center-3&pf_rd_e=gxwy42FH2K03Y7BMWQNM&pf_rd_t=101&pf_rd_p=1328901542&pf_rd_i=16579



Analyzing Requirements

- **Analyzing** requirements is the process of examining and understanding the requirements of a system.
 - It involves identifying dependencies, conflicts, and inconsistencies in the requirements.
 - It is a critical activity that ensures that the requirements are complete, consistent, and correct.
- System*



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Handwritten red annotations:

- A bracket under "critical activity" with an arrow pointing to "understanding".
- An arrow from "ensures" pointing to "feasible".



Documenting Requirements

classical
tech

modern (agile)
non-tech

- Documenting requirements is the process of writing and organizing the requirements of a system.
- It involves creating documents, diagrams, and models that describe the requirements in a clear and concise way.

• Docx

• md

- Wiki

↓
Atlassian

→ Jira
Bitbucket
Confluence

→ mockups
→ validate
→ tech tasks
processes
→ business

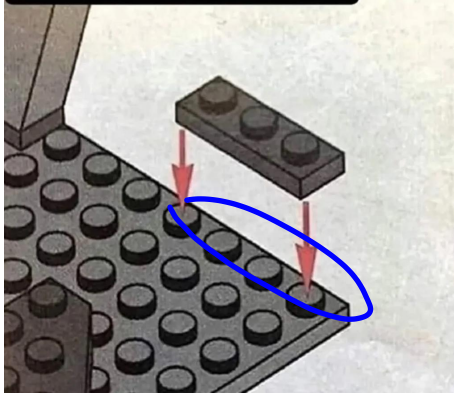
- priorities
- assumptions



Everyone hates to write Documentation

**En la documentación
está todo bien explicado**

La documentación:



Validating Requirements

- **Validating** requirements is the **process** of **ensuring** that the requirements are **correct** and **complete**.
- It involves **reviewing** the requirements with stakeholders to **verify** that they **meet** their **needs**.

meetings



negotiation

models
diagrams



solution



NOT Clear Understanding of Requirements



Dad Jokes
@Dadsaysjokes

..

My dad told me his password is:
MickeyMinnieGoofyDonaldPlutoHuey
LouieDeweyDublin.

Because he was told his password
had to contain 8 characters and at
least one Capital.

*NOT
Expectation*



Verifying Requirements → some sprints

- **Verifying** requirements is the process of ensuring that the requirements are correctly implemented in the system.
- It involves testing the system to verify that it meets the requirements.
- It is a critical activity that ensures that the system meets the needs of its users.

Testing engineering



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Typical Mistakes when Testing

Disturbing Chinese calorie app...

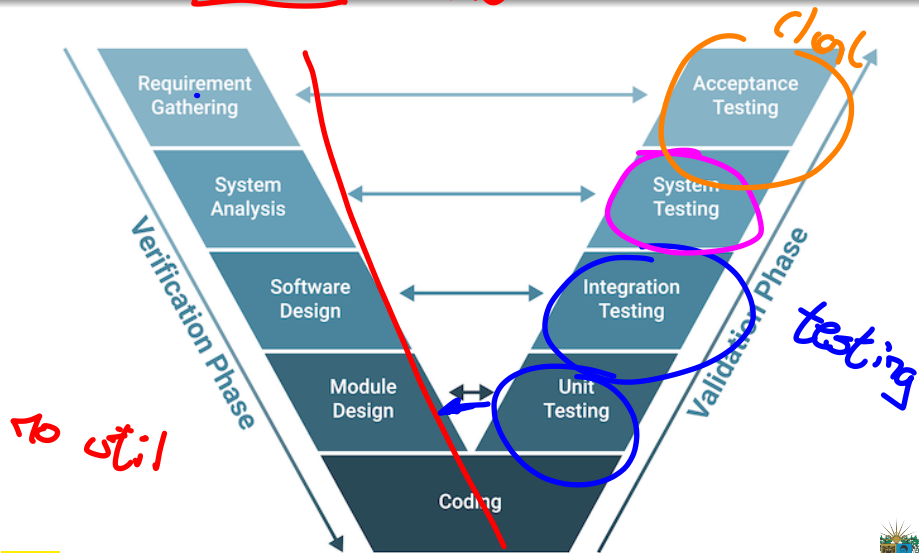


	カシューナッツ (cashew)	1 粒	9 kcal
	ジャムパン (Pomeranian)	1 個	327 kcal



V-Model in SDLC

1-1975~



Outline

1 Requirements Engineering

2 Design & Process



Conceptual Design

- Once the initial set of requirements is defined, the next step is to create a conceptual design of the system.
 - Conceptual Design** is a high-level design that defines the structure and behavior of the system. It is achieved by the recognition of the appropriate components, connections, and responsibilities. *Non-technical*
 - The conceptual design is used to communicate the vision of the system to stakeholders and to guide the development of the system. *all can understand.*
- system*



Conceptual Design

- Once the initial set of **requirements** is defined, the next step is to create a **conceptual design** of the system.
- **Conceptual Design** is a **high-level design** that defines the structure and behavior of the system. It is achieved by the recognition of the appropriate **components**, **connections**, and **responsibilities**.
- The **conceptual design** is used to **communicate** the **vision** of the system to **stakeholders** and to **guide** the **development** of the system.



Process Definition

- A **Process** is a **series** of steps or actions taken to achieve a particular end.
- **Processes** are used to **organize** and **manage** work.



Workflows

→ Pipeline

- A **Workflow** is a series of tasks that are performed in a specific order to achieve a goal.
- **Workflows** are used to automate and optimize business processes.
- **Workflows** can be sequential, parallel, conditional, or repetitive.

→
→
→
→
Flowchart



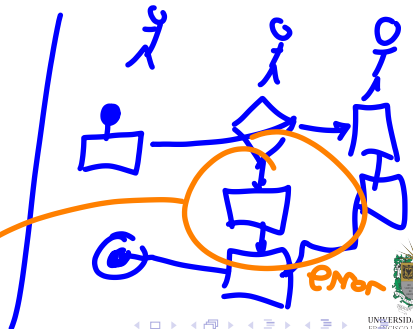
Process Models

dynamical system

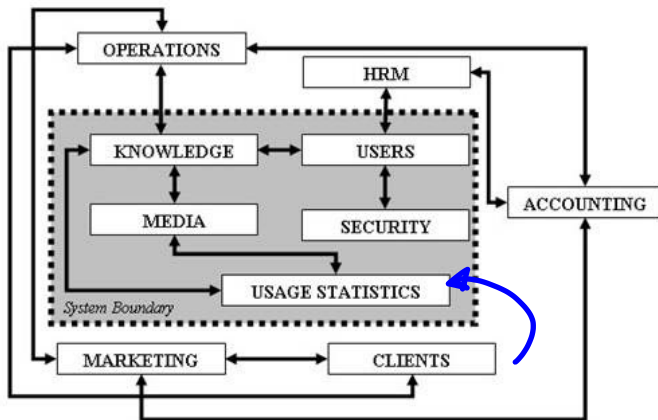
- A **Process Model** is a representation of a **process** that shows the sequence of steps and the **relationships** between them.
- **Process models** are used to **analyze**, **design**, and **improve** processes.
- Examples of **process models** include flowcharts, data flow diagrams, activity diagrams, **business process model and notation** (BPMN), petri nets, state diagrams, among others.

BPMN
activity
sequence

*new
project*



System Schema Example: Company Structure



Causal Loops

- A **Causal Loop** is a **diagram** that shows the **relationships** between different variables in a system.
- Causal loops are used to **analyze** and **understand** the **dynamics** of a system.
- Causal loops can be **positive** or **negative**.

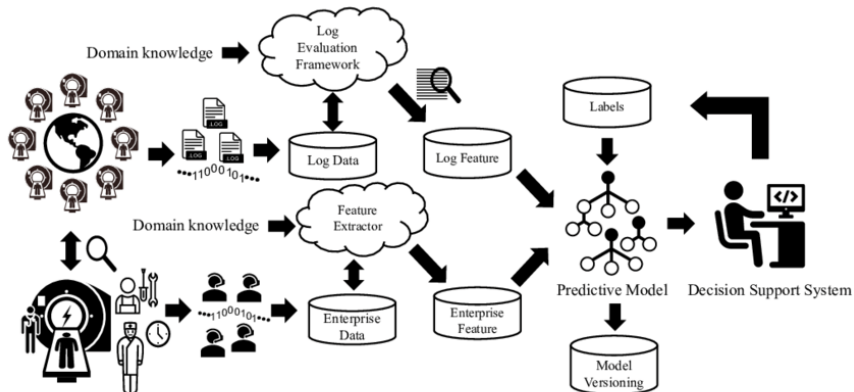


Causal Loops

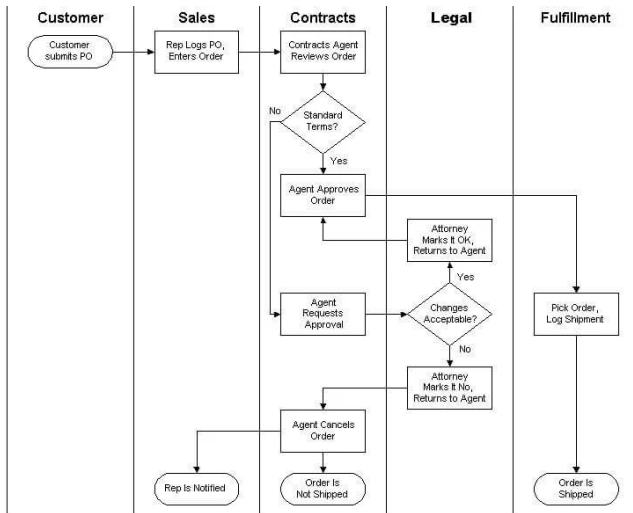
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System Schema Example: Processing Pipeline



Business Process Model and Notation (BPMN)



Technical Design

- Once the **conceptual design** of the system is defined, the next step is to create a **technical design** of the system.
- **Technical Design** is a **detailed design** that defines the architecture, components, and interfaces of the system.
- The **technical design** is used to **guide the development** of the system and to **communicate the implementation details** to **developers**.

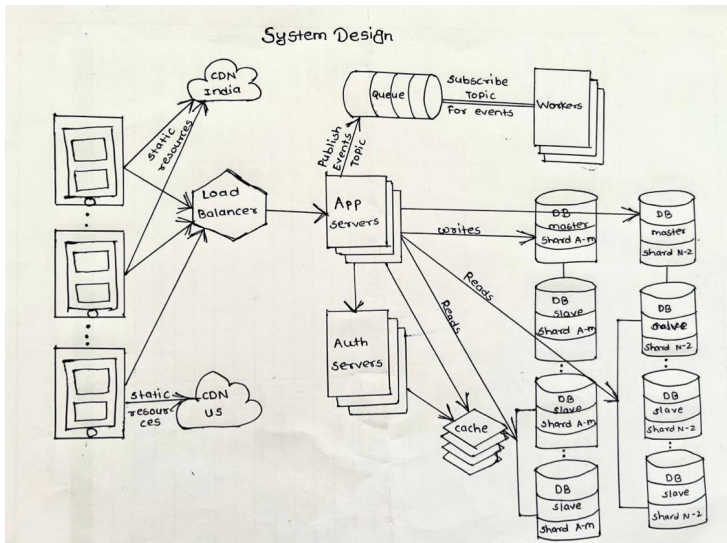


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Systems Design applied to Software Architectures



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

